



Engineering Materials

Combines principles from materials science and engineering to study the structure, properties, processing, and performance of materials used in engineering applications.

Common Types of Engineering Materials

Metals

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- Commonly used engineering materials due to their excellent strength, durability, and conductivity.
- Properties can be modified by alloying and heat treatment.
- Examples include steel, aluminum, copper, titanium, and nickel.
- Used in applications such as structural components, electrical wiring, and heat exchangers.

Polymers

- Large molecules made up of repeating subunits.
- Lightweight, versatile, and can have excellent chemical resistance.
- Can be classified as thermoplastics or thermosetting plastics.
- Widely used in packaging, automotive parts, electronics, and medical devices.



Ceramics

- Non-metallic, inorganic materials.
- Generally brittle but can have excellent heat resistance and electrical insulation properties.



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- Examples include glass, porcelain, alumina, and silicon carbide.
- Used in applications such as electronic components, cutting tools, and heat shields.

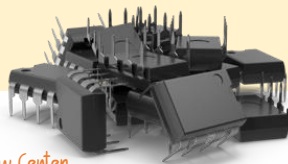
Composites

- Combination of two or more different materials to achieve desired properties.
- Consist of a matrix material and reinforcement fibers.
- Can have high strength-to-weight ratios and tailored properties.
- Examples include carbon fiber reinforced polymers (CFRP) and fiberglass
- Used in aerospace, automotive, and sporting goods industries.



Semiconductors

- Materials with electrical conductivity between conductors and insulators.
- Important for electronic devices and integrated circuits.
- Examples include silicon and gallium arsenide.
- Properties can be altered through doping to control electrical behavior.



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Biomaterials

- Materials used in medical and biological applications.
- Should be biocompatible, capable of interacting with living tissues without causing adverse reactions.
- Examples include metals, ceramics, polymers, and composites used in implants, prosthetics, and drug delivery systems.

What are alloys?

Alloys are metallic materials that are composed of a combination of two or more elements, at least one of which is a metal.

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Alloys are created to enhance specific properties of the individual metals, such as strength, hardness, corrosion resistance, or conductivity.

Steel

Steel is an alloy of iron and carbon, with carbon content typically ranging from 0.2% to 2.1% by weight. It is one of the most widely used alloys and offers excellent strength, durability, and versatility. Various alloying elements such as chromium, nickel, and manganese can be added to steel to enhance its properties further.



Bronze

Bronze is an alloy primarily composed of copper and tin. It has been used for thousands of years and is known for its strength, corrosion resistance, and attractive appearance.



Brass

Brass is an alloy of copper and zinc, with varying compositions depending on the desired properties. It is valued for its excellent malleability, corrosion resistance, and acoustic properties.



Stainless Steel

Stainless steel is an alloy of iron, chromium, and often other elements such as nickel and molybdenum. It is highly resistant to corrosion.

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